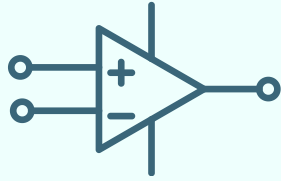


FUNDAMENTALS OF MIXED SIGNALS & SENSORS

Dave Jhared G. Paduada
BSCpE III GF

OPERATIONAL AMPLIFIER



- or **op-amp**, is a very high gain differential amplifier with high input impedance and low output impedance.
- is basically a multistage, very high gain (typically 200 000), direct coupled, negative feedback amplifier that provides a stabilized voltage gain.
- An op-amp contains a number of differential amplifier stages to achieve a very high voltage gain.



OPERATIONAL AMPLIFIER CHARACTERISTICS

Very high gain differential amplifier
High input impedance (M- Ω)
Low output impedance ($< 100 \Omega$)



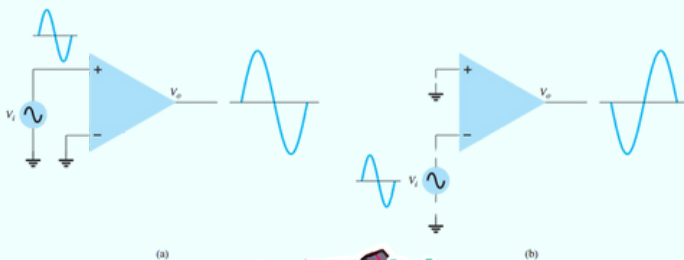
OP-AMPS APPLICATIONS

Voltage amplitude and polarity changes
Oscillators
Filter Circuits
Instrumentation Circuits

Single Ended
Double Ended

Inputs & Outputs

Single Ended
Double Ended



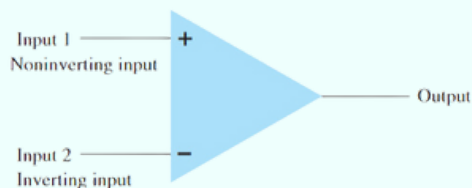
Single Ended



Double ended

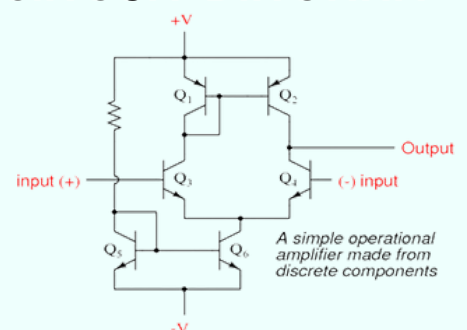


Basic OP-amp symbol

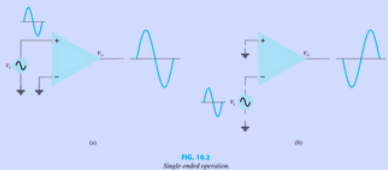


Plus (+) input: in phase output
Minus (-) input: opposite polarity output

OP-AMP EQUIVALENT CIRCUIT DIAGRAM

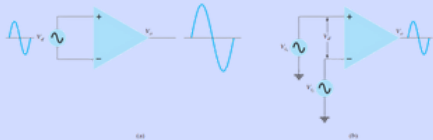


SINGLE ENDED INPUT



- Single-ended input operation
- Input signal
- One input connected
- Other input connected to ground

DOUBLE ENDED INPUT



Two separate signals are applied to the inputs, the difference signal being:

$V_1 - V_2$



DOUBLE ENDED OUTPUT

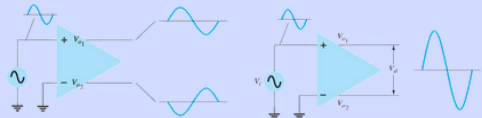


FIG. 10.5 Single-ended input with double-ended output.

$$V_d = V_{o1} - V_{o2} = V_{o1} - (-V_{o1}) = 2V_{o1}$$
$$= -V_{o2} - V_{o2} = -2V_{o2}$$

- Input: between two input terminals
- Output: between two output terminals
- Fully differential operation

COMMON MODE OPERATION

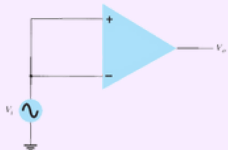


FIG. 10.8 Common mode operation.

- Same input signals: common mode operation
- Two inputs: equally amplified
- Opposite polarity signals: cancel
- Ideal output: 0 V
- Practical output: small signal

COMMON MODE INPUT

$V_C = 1/2 (V_{i1} + V_{i2})$



NOISE



(any unwanted input signal) is generally common to both inputs, the differential connection tends to provide attenuation of this unwanted input while providing an amplified output of the difference signal applied to the inputs.

This operating feature is referred to as **common - mode rejection**.

DIFFERENTIAL

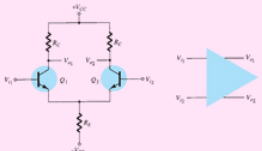


FIG. 10.9 Basic differential amplifier circuit.

ability to greatly amplify signals that are opposite at the two inputs, while only slightly amplifying signals that are common to both inputs

DIFFERENTIAL INPUTS

$V_d \approx V_{i1} - V_{i2}$

When separate inputs are applied to the op-amp, the resulting difference signal is the difference between the two inputs.

OUTPUT VOLTAGE

Since any signals applied to an op-amp in general have both in-phase and out-of-phase components, the resulting output can be expressed as

$V_o = A_d V_d + A_c V_c$

Where:

- V_d = difference voltage
- V_c = common voltage
- A_d = differential gain of the amp
- A_c = common mode gain of the amp

OPPOSITE POLARITY INPUTS

If opposite polarity inputs applied to an op-amp are ideally opposite signals

$V_{i1} = -V_{i2} = V_s$

resulting difference voltage:

$V_d = V_{i1} - V_{i2} = V_s - (-V_s) = 2V_s$

while the resulting common voltage is:

$V_c = \frac{1}{2}(V_{i1} + V_{i2}) = \frac{1}{2}[V_s + (-V_s)] = 0$

so that the resulting output voltage is

$$V_o = A_d V_d + A_c V_c$$
$$= A_d(2V_s) + 0$$
$$= 2A_d V_s$$

When the inputs are an ideal opposite signal (no common element), the output is the differential gain times twice the input signal applied to one of the inputs.

SAME POLARITY INPUTS

If opposite polarity inputs applied to an op-amp are ideally opposite signals

$V_{i1} = V_{i2} = V_s$

the resulting difference voltage is

$V_d = V_{i1} - V_{i2} = V_s - V_s = 0$

while the resulting common voltage is

$V_c = \frac{1}{2}(V_{i1} + V_{i2}) = \frac{1}{2}(V_s + V_s) = V_s$

so that the resulting output voltage is:

$$V_o = A_d V_d + A_c V_c$$
$$= A_d(0) + A_c V_s = A_c V_s$$

When the inputs are ideal in-phase signals (no difference signal), the output is the common-mode gain times the input signal, V_s , which shows that only common-mode operation occurs.

COMMON MODE REJECTION

AD:
set $V_{i1} = -V_{i2} = V_s = 0.5 \text{ V}$

$V_d = (V_{i1} - V_{i2}) = [0.5 \text{ V} - (-0.5 \text{ V})] = 1 \text{ V}$
 $V_c = \frac{1}{2}(V_{i1} + V_{i2}) = \frac{1}{2}[0.5 \text{ V} + (-0.5 \text{ V})] = 0 \text{ V}$

By substitution, the output voltage is

$V_o = A_d V_d + A_c V_c = A_d(1 \text{ V}) + A_c(0) = A_d$
 $V_{i1} = -V_{i2} = V_s = 0.5 \text{ V}$

AC:

$V_{i1} = V_{i2} = V_s = 1 \text{ V}$

$V_d = (V_{i1} - V_{i2}) = (1 \text{ V} - 1 \text{ V}) = 0 \text{ V}$
 $V_c = \frac{1}{2}(V_{i1} + V_{i2}) = \frac{1}{2}(1 \text{ V} + 1 \text{ V}) = 1 \text{ V}$

By substitution, the output voltage is

$V_o = A_d V_d + A_c V_c$
$$= A_d(0 \text{ V}) + A_c(1 \text{ V}) = A_c$$

$V_{i1} = V_{i2} = V_s = 1 \text{ V}$

COMMON MODE REJECTION - RATIO

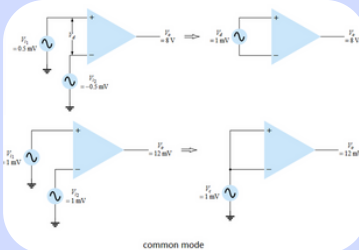
Having obtained A_d and A_c , we can now calculate a value for the common-mode rejection ratio (CMRR), which is

$$CMRR = \frac{A_d}{A_c}$$

The value of CMRR can also be expressed in logarithmic terms as:

$$CMRR (\log) = 20 \log_{10} \frac{A_d}{A_c}$$

SAMPLE PROBLEM



for A_d : $A_d = \frac{V_o}{V_d} = \frac{8 \text{ V}}{1 \text{ mV}} = 8000$

for A_c : $A_c = \frac{V_o}{V_c} = \frac{12 \text{ mV}}{1 \text{ mV}} = 12$

$CMRR = \frac{A_d}{A_c} = \frac{8000}{12} = 666.7$

$CMRR (\log) = 20 \log_{10} \frac{A_d}{A_c} = 20 \log_{10} 666.7 = 56.48 \text{ dB}$

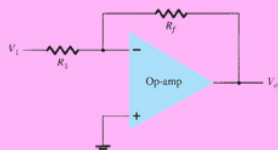
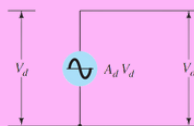
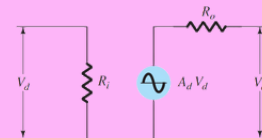


FIG. 10.31
Basic op-amp connection.

- Basic circuit: op-amp connection
- Constant-gain multiplier
- Input signal V1: through resistor R1 to minus input
- Output: connected to minus input through resistor Rf
- Plus input: connected to ground
- Output: opposite phase to input signal



An ideal op-amp circuit, would have infinite input impedance, zero output impedance, and an infinite voltage gain.



The input signal applied between input terminals sees an input impedance, Ri, typically very high. The output voltage is shown to be the amplifier gain times the input signal taken through an output impedance, Ro, which is typically very low.

PRACTICAL OP-AMP CIRCUITS

INVERTING AMP

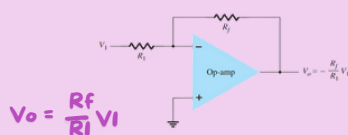


FIG. 10.34
Inverting constant-gain multiplier.

most widely used constant-gain amplifier circuit.

Example: If the circuit of Fig. 10.34 has R1 = 100 kΩ and Rf = 500kΩ, what output voltage results for an input of V1 = 2 V?

$$V_o = -\frac{R_f}{R_1} V_1 = -\frac{500 \text{ k}\Omega}{100 \text{ k}\Omega} (2 \text{ V}) = -10 \text{ V}$$

NON-INVERTING AMP

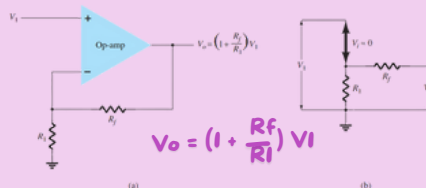


FIG. 10.35
Noninverting constant-gain multiplier.

connection is more widely used because it has better frequency stability.

Problem: Calculate the output voltage of a noninverting amplifier (as in Fig. 10.35) for values of V1 = 2 V, Rf = 500 kΩ, and R1 = 100 kΩ.

$$\begin{aligned} V_o &= \left(1 + \frac{R_f}{R_1}\right) V_1 \\ &= \left(1 + \frac{500 \text{ k}\Omega}{100 \text{ k}\Omega}\right) (2 \text{ V}) = 6(2 \text{ V}) \\ &= +12 \text{ V} \end{aligned}$$

UNITY FOLLOWER

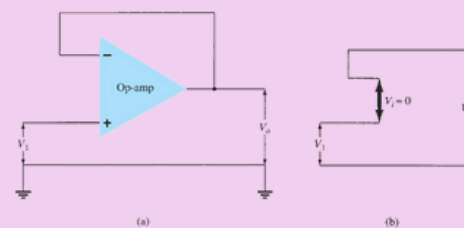


FIG. 10.36
(a) Unity follower; (b) virtual-ground equivalent circuit.

$$V_o = V_i$$

The unity-follower circuit provides a gain of unity with no polarity or phase reversal. The output is the same polarity and magnitude as the input. The circuit operates like an emitter- or source-follower circuit except that the gain is exactly unity.

SUMMING AMP

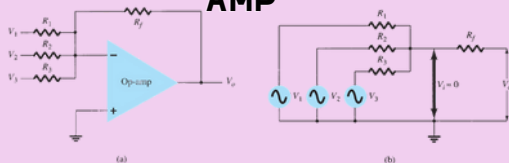


FIG. 10.37
(a) Summing amplifier; (b) virtual-ground equivalent circuit.

$$V_o = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3\right)$$

Most used of the op-amp circuits is the summing amplifier circuit. The circuit shows a three-input summing amplifier circuit, which provides a means of algebraically summing (adding) three voltages, each multiplied by a constant-gain factor.

Problem: Calculate the output voltage of an op-amp summing amplifier for the following sets of voltages and resistors. Use Rf = 1 MΩ in all cases.

Solution: Using Eq. (10.11), we obtain

$$\begin{aligned} \text{a. } V_o &= -\left[\frac{1000 \text{ k}\Omega}{500 \text{ k}\Omega} (+1 \text{ V}) + \frac{1000 \text{ k}\Omega}{1000 \text{ k}\Omega} (+2 \text{ V}) + \frac{1000 \text{ k}\Omega}{1000 \text{ k}\Omega} (+3 \text{ V})\right] \\ &= -[2(1 \text{ V}) + 1(2 \text{ V}) + 1(3 \text{ V})] = -7 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{b. } V_o &= -\left[\frac{1000 \text{ k}\Omega}{200 \text{ k}\Omega} (-2 \text{ V}) + \frac{1000 \text{ k}\Omega}{500 \text{ k}\Omega} (+3 \text{ V}) + \frac{1000 \text{ k}\Omega}{1000 \text{ k}\Omega} (+1 \text{ V})\right] \\ &= -[5(-2 \text{ V}) + 2(3 \text{ V}) + 1(1 \text{ V})] = +3 \text{ V} \end{aligned}$$

Integrator

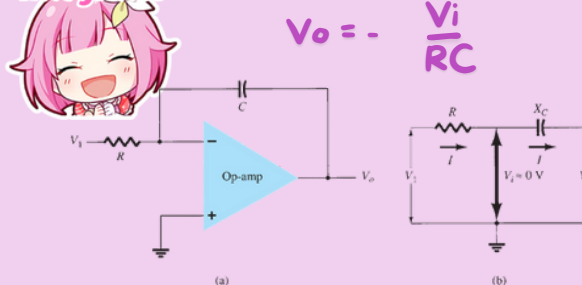
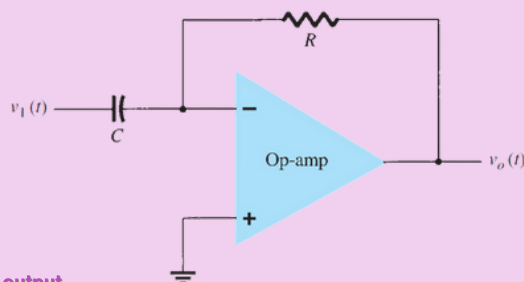


FIG. 10.38
Integrator.

If the feedback component used is a capacitor, the result is an integrator circuit. The integrator produces a voltage output proportional to the product (multiplication) of the input voltage and time.

Differentiator



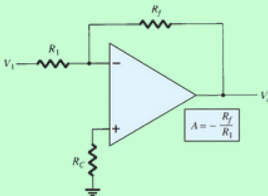
$$V_o = -RC \frac{dV_i}{dt}$$

produces a voltage output proportional to the input voltage's rate of change.

FIG. 10.41
Differentiator circuit.

OP-AMP APPLICATIONS

INVERTING FIXED GAIN AMP



One of the most common op-amp circuits is the inverting constant-gain multiplier, which provides a precise gain or amplification.

Problem: Determine the output voltage for the circuit of Fig. 11.2 with a sinusoidal input of 2.5 mV.

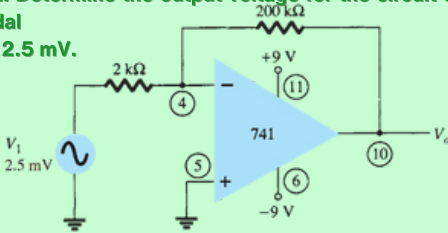


FIG. 11.2
Circuit for Example 11.1.

Solution: The circuit of Fig. 11.2 uses a 741 op-amp to provide a constant or fixed gain, calculated from Eq. (11.1) to be

$$A = -\frac{R_f}{R_1} = -\frac{200 \text{ k}\Omega}{2 \text{ k}\Omega} = -100$$

The output voltage is then

$$V_o = AV_i = -100(2.5 \text{ mV}) = -250 \text{ mV} = -0.25 \text{ V}$$

NON-INVERTING FIXED GAIN AMP

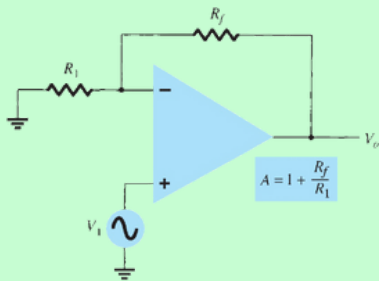


FIG. 11.3
Noninverting fixed-gain amplifier.

$$A = 1 + \frac{R_f}{R_1}$$

VOLTAGE SUBTRACTION

Two signals can be subtracted from one another in a number of ways. Figure 11.10 shows two op-amp stages used to provide subtraction of input signals. The resulting output is given by

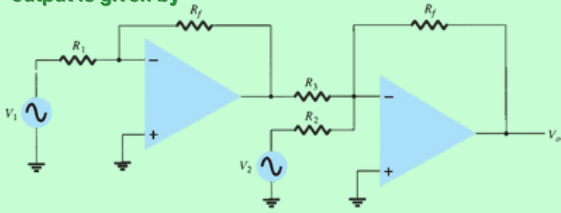


FIG. 11.10
Circuit for subtracting two signals.

$$V_o = -\left(\frac{R_f}{R_2}V_2 - \frac{R_f}{R_3}\frac{R_f}{R_1}V_1\right)$$



Another connection to provide subtraction of two signals uses only one op-amp stage to provide subtracting two input signals. Using superposition the output can be shown to be:

$$V_o = \frac{R_3}{R_1 + R_3} \frac{R_2 + R_4}{R_2} V_1 - \frac{R_4}{R_2} V_2$$

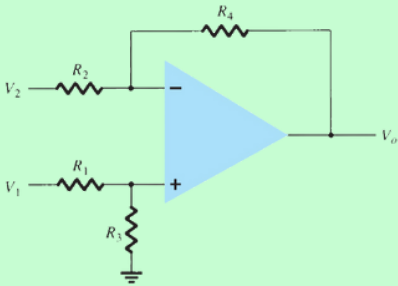


FIG. 11.11
Subtraction circuit.

MULTIPLE-STAGE GAINS

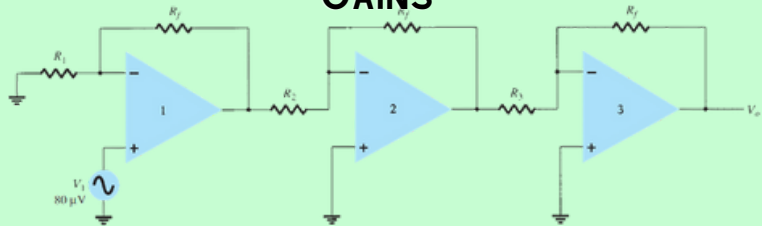


FIG. 11.5
Constant-gain connection with multiple stages.

When a number of stages are connected in series, the overall gain is the product of the individual stage gains.

EXAMPLE 11.3 Calculate the output voltage using the circuit of Fig. 11.5 for resistor components of value $R_f = 470 \text{ k}\Omega$, $R_1 = 4.3 \text{ k}\Omega$, $R_2 = 33 \text{ k}\Omega$, and $R_3 = 33 \text{ k}\Omega$ for an input of $80 \mu\text{V}$.

Solution: The amplifier gain is calculated to be

$$\begin{aligned} A &= A_1 A_2 A_3 = \left(1 + \frac{R_f}{R_1}\right) \left(-\frac{R_f}{R_2}\right) \left(-\frac{R_f}{R_3}\right) \\ &= \left(1 + \frac{470 \text{ k}\Omega}{4.3 \text{ k}\Omega}\right) \left(-\frac{470 \text{ k}\Omega}{33 \text{ k}\Omega}\right) \left(-\frac{470 \text{ k}\Omega}{33 \text{ k}\Omega}\right) \\ &= (110.3)(-14.2)(-14.2) = 22.2 \times 10^3 \end{aligned}$$

so that

$$V_o = AV_i = 22.2 \times 10^3(80 \mu\text{V}) = 1.78 \text{ V}$$

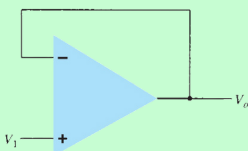


FIG. 11.13
Unity-gain (buffer) amplifier.

A voltage buffer circuit provides a means of isolating an input signal from a load by using a stage having unity voltage gain, with no phase or polarity inversion, and acting as an ideal circuit with very high input impedance and low output impedance.

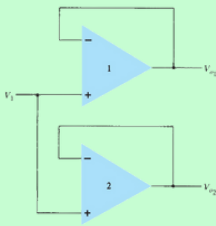
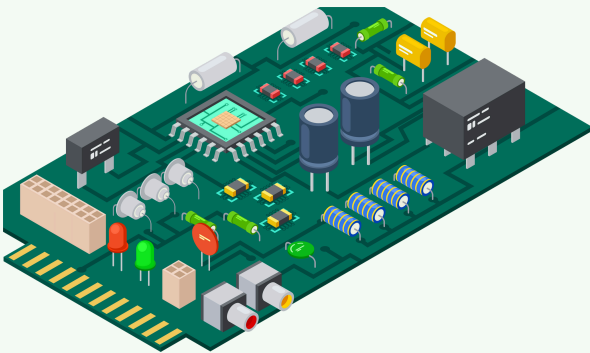


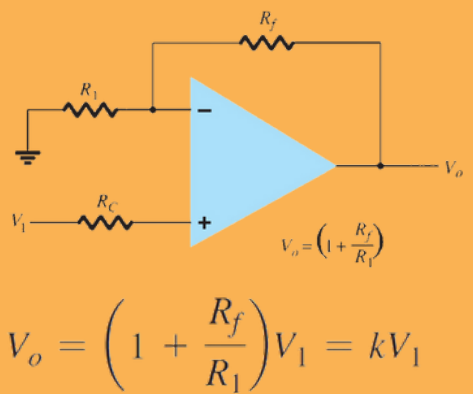
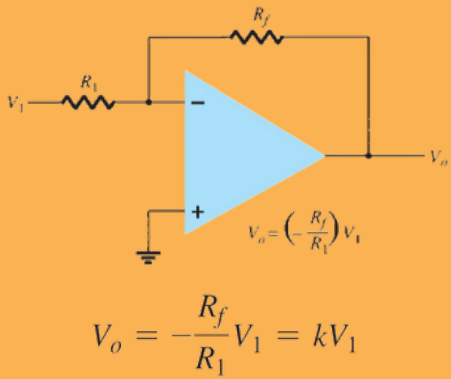
FIG. 11.14
Use of buffer amplifier to provide output signals.

The advantage of this connection is that the load connected across one output has no or little effect on the other output. The outputs are buffered or isolated from each other.



CONTROLLED SOURCES

VOLTAGE-CONTROLLED VOLTAGE SOURCE



Two versions of the circuit are shown, one using the inverting input, the other the non-inverting input. An ideal form of a voltage source whose output V_o is controlled by an input voltage V_1 .

The output voltage is seen to be dependent on the input voltage (times a scale factor k).

VOLTAGE-CONTROLLED CURRENT SOURCE

An ideal form of circuit providing an output current controlled by an input voltage. The output current is dependent on the input voltage.

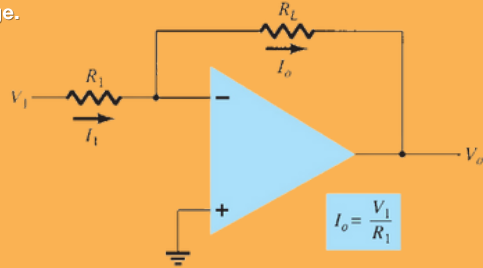


FIG. 11.19

Practical voltage-controlled current source.

$$I_o = \frac{V_1}{R_1} = kV_1$$

CURRENT-CONTROLLED VOLTAGE SOURCE

The output voltage is dependent on the input current.

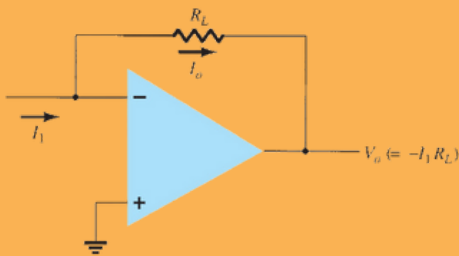


FIG. 11.21

Practical form of current-controlled voltage source.

$$V_o = -I_1R_L = kI_1$$

CURRENT-CONTROLLED CURRENT SOURCE

An output current is provided dependent on the input current.

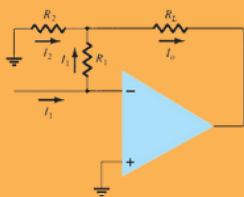


FIG. 11.23

Practical form of current-controlled current source.

$$I_o = I_1 + I_2 = I_1 + \frac{I_1R_1}{R_2} = \left(1 + \frac{R_1}{R_2}\right)I_1 = kI_1$$

EXAMPLE 11.10

- For the circuit of Fig. 11.24a, calculate I_L .
- For the circuit of Fig. 11.24b, calculate V_o .

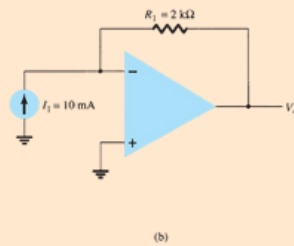
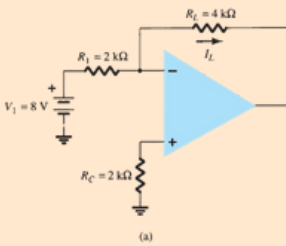


FIG. 11.24

Circuits for Example 11.10.

Solution:

- For the circuit of Fig. 11.24a,

$$I_L = \frac{V_1}{R_1} = \frac{8 \text{ V}}{2 \text{ k}\Omega} = 4 \text{ mA}$$

- For the circuit of Fig. 11.24b,

$$V_o = -I_1R_1 = -(10 \text{ mA})(2 \text{ k}\Omega) = -20 \text{ V}$$



My exam:



